1. Objective:

Understand programming and the mode of operation of programmable logic devices.

For this purpose a schematic similar to that from lab exercise 2 will be build using a board containing an Altera FPGA and some additional logic on an additional board.

The boards are given, only the FPGA has to be programmed accordingly.

2. Preparation:

2.1. An Overview of the boards and additional relevant documents can be found in the ILIAS folder “LAB_FPGA_Docs”.

2.2. A schematic for the Altera (Quartus II) software package is available in the ILIAS (Lab_03_Schematic.pdf). It differs from the task from lab exercise 2 in the following:

- 2.1.1. LED0 … LED7 are used to display the “light”. Notice that these LEDs are external to the Altera FPGA and are located on the FPGA board.
- 2.1.2. COUNT0 … COUNT3 are used to display the count status. Notice that these LEDs are external to the Altera FPGA and are located on the additional board. Also note that these LEDs are “ON” when driven by a low logic level!
- 2.1.3. Push-button Key0 is used for reset. Push-buttons are external to the FPGA and are located on the FPGA board. The pushbuttons are debounced.
- 2.1.4. Push-button Key1 is used to “catch” the light.
- 2.1.5. The frequency generation for the moving light is done internally in the FPGA. It has a period of approx. 1 second.
- 2.1.6. For “catching” the light, use the NE555 available on the additional board. The time delay is approx. 2 seconds. Use the “NE555-TRIGGER” output of the FPGA to start the monostable multivibrator and the “NE555-OUT” as input from it.

2.3. Print the PDF file and analyze the schematic given.

2.4. Analyze the block schematic of the Altera FPGA board and the schematic of the additional board.

2.5. Design the missing logic for the locations indicated by: ???

2.6. The schematic has to be presented to the tutor before starting the lab exercise.

3. Tasks:

3.1. Unpack the files contained in the ZIP (Lab3_Project.zip) file into a directory of your choice and use the Altera (Quartus II) software package to open the “Quartus project file” (*.qpf).

3.2. Open the schematic file and add the missing logic at the locations indicated by: ???

3.3. Note that pin numbers are already assigned to the inputs and outputs!

3.4. Compile the project, program it onto the board and make it run.

3.5. Consider the spiral design model!

4. Report:

Not requested.

5. Notes:

None